

Arithmetic from Distinction: A Number Tower Forced by a Single Primitive

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May 2026

Abstract

We construct the natural numbers, the integers, and the rational field from a single primitive: the act of distinction. The construction uses no axiom of infinity in the set-theoretic sense, no axiom of choice, and no law of excluded middle. Each object is a finite record of performed distinctions, and every theorem is proved by structural induction over such finite records. From the primitive we build finite traces, a sameness relation that certifies the absence of a separating distinction, a difference relation that certifies its presence, and a quotient discipline under which any admissible predicate must respect sameness. Counting is the length of a distinction orbit. Sign is the balance of two orbit lengths. Ratio is the cross-comparison of two signed orbits against two lengths. On these objects we prove the full slate of elementary arithmetic: associativity, commutativity, distributivity, cancellation, a decidable total order, absolute value, divisibility, primality, Euclidean division with unique quotient and remainder, greatest common divisors, coprimality, and the field axioms for the rationals. Each construction carries a proof-strength tag, and every result in this paper carries the strongest tag available: it is forced by distinction and finite repetition alone. To certify that the construction recovers the familiar objects and nothing stranger, we exhibit faithful display maps onto the standard naturals, integers, and rationals, and we prove these maps are isomorphisms of the relevant algebraic structures. The display maps are verification instruments, not ingredients: the orbit operations are defined without reference to any prior numeral system. The upshot is that the elementary number tower is not a convention chosen on top of a foundation but a structure that the single act of distinction permits and then forces. We close by marking precisely where this forcing stops. The completion of the rational field into the continuum is a separate commitment, treated in a companion paper, and it is not a consequence of distinction.

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1 What this paper does

Foundations of mathematics usually begin with a rich primitive and then encode number inside it. Set theory begins with membership and builds the number three as a particular nest of braces. Type theory begins with types and terms and introduces the natural numbers as a particular inductive type. Both encodings work, and both are correct. Both also carry a feature that is easy to overlook because it is so familiar: the primitive has nothing to do with counting. A set is not a number. A type is not a number. The number arrives only after a chain of definitions that translate the foreign primitive into arithmetic behavior.

This paper takes a different primitive. We begin with the act of distinction, the operation of marking that one thing is not another, and we show that the elementary number tower follows from it directly. Counting is the most natural reading of repeated distinction: to count is to perform the same act of marking again and again and to keep a record of how many times. Sign is the most natural reading of two opposed directions of counting. Ratio is the most natural reading of comparing two counts. None of these is imposed as a definitional convention. Each is the structure that the primitive already contains once you take it seriously.

The reason to care is not that we obtain a new arithmetic. The integers built here are isomorphic to the usual integers, and the rationals to the usual rationals; we prove exactly that. The reason to care is the audit. Every result in this paper is established with a single, fixed, minimal proof budget, and we keep an explicit ledger of that budget. We use no axiom of infinity in the set-theoretic sense, meaning we never posit a completed infinite set; we work only with finite records and the two rules that generate them. We use no axiom of choice, because every object we need is constructed explicitly rather than selected from a family. We use no law of excluded middle, because every predicate we rely on is decidable by a terminating procedure on finite data. A foundation that announces a single primitive owes the reader a precise account of what that primitive can and cannot reach without help. This paper is that account for the arithmetic layer.

1.1 Contributions

The paper makes four contributions.

1. It defines a finite kernel from the primitive act of distinction: traces, trace concatenation with its associativity law, the sameness and difference relations with their consistency condition, and the quotient discipline that any admissible predicate must obey.
2. It constructs the natural numbers as distinction orbits and proves the full arithmetic slate for them, including a decidable total order and a cancellation law, by structural induction alone.

3. It constructs the integers as balanced pairs of orbit lengths and the rationals as cross-compared ratios, and it proves that these carry a commutative ring structure and an ordered field structure respectively, each isomorphic to the standard object under a faithful display map.
4. It derives the elementary number-theoretic surface, divisibility, primality, Euclidean division, greatest common divisors, and coprimality, with the same minimal proof budget, and it states exactly where the budget would have to grow, namely at the completion into the continuum, which is deferred to a companion paper.

1.2 Status discipline

A paper that claims to derive arithmetic from one act, rather than to encode it inside a richer theory, must be careful about what counts as derivation and what smuggles in a hidden assumption. We adopt a per-theorem discipline. Every construction is labeled with a proof-strength tag, defined in Section 4, and every result in this paper carries the same tag: it is forced by distinction and finite repetition alone, with constructive logic and no further commitments. When we use a display map into a standard object, we say so explicitly and we explain that the map is used to certify faithfulness after the fact, not to define the construction. The reader should be able to check at every step that nothing has entered the argument except the primitive and induction over finite records.

2 The primitive: distinction acts and traces

The primitive is an act, not an object. It is the operation of marking a difference, taken in its most contentless form.

Definition 2.1 (Distinction act). A *distinction act* is a single, atomic operation of marking a difference. It carries no parameters, no payload, and no internal structure. It is pure separation: the minimal operation that takes an undifferentiated situation to a differentiated one. We write δ for this act.

A distinction act on its own is too thin to support mathematics. What supports mathematics is the record of having performed distinction acts, possibly many times. That record is a trace.

Definition 2.2 (Trace). A *trace* is a finite sequence of distinction acts, defined by two formation rules:

- (i) the *empty trace* ε is a trace, recording that no distinction has been performed;
- (ii) if τ is a trace, then $\tau \cdot \delta$, the result of extending τ by one further distinction act, is a trace.

The *length* of a trace is the number of extension steps used to build it: $|\varepsilon| = 0$ and $|\tau \cdot \delta| = |\tau| + 1$. The *concatenation* $\tau_1 \parallel \tau_2$ of two traces is defined by recursion on the second argument:

$$\tau_1 \parallel \varepsilon = \tau_1, \quad \tau_1 \parallel (\tau_2 \cdot \delta) = (\tau_1 \parallel \tau_2) \cdot \delta.$$

Two features of this definition deserve emphasis, because they carry the whole budget claim of the paper. First, every trace is finite. There is no rule that produces an infinite trace, and the two rules above generate only objects built from finitely many extension steps. When we later quantify over all traces, we quantify over a collection generated by these rules, exactly as one quantifies over the natural numbers generated by zero and successor. We never posit a completed infinite totality as a new object with its own axiom. Second, the primitive is an act, not a set, a type, a proposition, or a field. Set theory tells you what a set is; type theory tells you what a type is; the present theory tells you what distinguishing is, and lets sets, types, and numbers appear later as constructions rather than as starting assumptions.

Lemma 2.3 (Concatenation is associative). *For all traces τ_1, τ_2, τ_3 ,*

$$(\tau_1 \parallel \tau_2) \parallel \tau_3 = \tau_1 \parallel (\tau_2 \parallel \tau_3).$$

Proof. By structural induction on τ_3 . If $\tau_3 = \varepsilon$, both sides equal $\tau_1 \parallel \tau_2$, using the base clause of concatenation twice. For the inductive step, suppose the identity holds for τ_3 and consider $\tau_3 \cdot \delta$. Using the recursive clause of concatenation on each side and the induction hypothesis,

$$\begin{aligned} (\tau_1 \parallel \tau_2) \parallel (\tau_3 \cdot \delta) &= ((\tau_1 \parallel \tau_2) \parallel \tau_3) \cdot \delta = (\tau_1 \parallel (\tau_2 \parallel \tau_3)) \cdot \delta \\ &= \tau_1 \parallel ((\tau_2 \parallel \tau_3) \cdot \delta) = \tau_1 \parallel (\tau_2 \parallel (\tau_3 \cdot \delta)). \end{aligned}$$

This is the claim for $\tau_3 \cdot \delta$, completing the induction. $\square$

3 Sameness, difference, and the quotient discipline

Traces let us record distinctions, but mathematics also needs the converse: a way to certify that two things have not been distinguished. In set theory, equality is a logical primitive fixed by extensionality. In type theory, it is the identity type. In a theory built on acts of distinction, the natural primitive is a certification relative to a trace: two endpoints are the same when no distinction in the current trace separates them, and different when one does.

Definition 3.1 (Sameness and difference). Let A be a collection of trace endpoints. We introduce two relations on A . The *sameness* relation $a \sim b$ asserts that no distinction in the current trace separates a from b . The *difference* relation $a \# b$ asserts that some distinction in the trace does separate them. We require $\sim$ to be an equivalence relation: reflexive, symmetric, and transitive. We require the *consistency condition*, that for any pair (a, b) and any fixed respect, sameness and difference cannot both hold,

$$a \sim b \wedge a \# b \implies \perp.$$

The relation $\sim$ is weaker than logical equality, and this is the point. Two endpoints can be the same relative to a coarse trace and different relative to a finer one. Sameness is a statement about what the present record can resolve, not a metaphysical claim about identity. This is the right primitive for a theory whose only generator is the act of distinction: you are entitled to treat two things as the same exactly when you have performed no act that tells them apart.

Once endpoints are identified under sameness, every construction built on top must respect that identification. Otherwise the construction would distinguish what the trace cannot, which violates the spirit of the primitive.

Definition 3.2 (Admissible predicate and quotient descent). A predicate P on endpoints is *admissible* if it respects sameness: whenever $a \sim b$, we have $P(a)$ if and only if $P(b)$. An admissible predicate descends to a well-defined predicate on the quotient $A/\sim$. More generally, a function f on representatives descends to a function on $A/\sim$ whenever $a \sim b$ implies $f(a)$ and $f(b)$ agree in the relevant sense. This descent principle is the only tool we use to move from representatives to quotient objects, and it is available because $\sim$ is an equivalence relation.

The kernel is now complete. It consists of finite traces with their concatenation and associativity law, the sameness and difference relations with their consistency condition, and the descent principle for admissible predicates and functions. Everything in the rest of the paper is built from this kernel by finite construction and structural induction.

Tag	Meaning
distinction-only	Uses only the finite kernel and constructive logic. No infinity, choice, or excluded middle.
distinction + closure	Adds completed infinite streams or limiting boundary commitments.
distinction + choice	Uses a choice principle to select representatives or witnesses.
classical extension	Uses classical logic or an imported classical carrier such as the real line.

Table 1: The proof-strength ledger. Each level includes the machinery of all levels below it. Every theorem in this paper carries the lowest tag, distinction-only.

4 The proof-strength ledger

Conventional mathematics answers the question “what are your assumptions?” once, globally, and usually with a large answer such as set theory together with choice and classical logic. We answer it per theorem, with a tag attached to each result. The tags form a lattice ordered by how much machinery each level adds beyond the bare kernel.

The discipline costs the author bookkeeping and gives the reader something most foundations do not provide: granular proof-strength information at the level of the individual theorem. A reader who cares only about constructive content can read the distinction-only results and stop. A reader who wants to know exactly where a completed infinity or a choice principle would first be required can find that boundary marked. In this paper the boundary is never crossed: the arithmetic tower through the rational field lives entirely at the distinction-only level. The first genuine crossing, into the continuum, is the subject of the companion paper and is not attempted here.

5 Base-neutral number

Number is the length of a record of repeated distinction. We call such a record a distinction orbit, and we treat the orbit itself as the number, prior to any numeral system used to display it.

Definition 5.1 (Distinction orbit, $\mathbb{N}_\delta$). The collection $\mathbb{N}_\delta$ of *distinction orbits* is generated by two rules:

- (i) the *zero orbit* 0_δ is an orbit, recording that no distinction has been performed;
- (ii) if n is an orbit, then its *successor* $S_\delta(n)$, recording one further distinction, is an orbit.

The *display map* $\varphi: \mathbb{N}_\delta \rightarrow \mathbb{N}$ onto the standard naturals is fixed by $\varphi(0_\delta) = 0$ and $\varphi(S_\delta(n)) = \varphi(n) + 1$.

The phrase *base-neutral* records that the orbit precedes every numeral system. The decimal symbol 3, the binary symbol 11, the Roman symbol III, and the successor expression $S_\delta(S_\delta(S_\delta(0_\delta)))$ are four display surfaces for one object: three performed distinctions. None of those symbols is the number; each is a way of writing it. The orbit is the number, and the display map converts it into whichever numeral system a given application finds convenient. Tally marks on a wall are closer to the orbit than the positional decimal system is, because tally marks are direct records of performed acts and carry no positional weighting, no carry rule, and no zero-padding. Those features belong to the numeral system, not to the number.

We define arithmetic by recursion on orbit structure, never referring to any numeral system.

Definition 5.2 (Orbit addition and multiplication). Addition $\oplus$ and multiplication $\otimes$ on $\mathbb{N}_\delta$ are defined by recursion on the second argument:

$$n \oplus 0_\delta = n, \quad n \oplus S_\delta(m) = S_\delta(n \oplus m); \quad (1)$$

$$n \otimes 0_\delta = 0_\delta, \quad n \otimes S_\delta(m) = (n \otimes m) \oplus n. \quad (2)$$

Addition is concatenation of orbits: appending m further distinctions after n . Multiplication is repeated addition: $n \otimes m$ adds n to itself m times. Both are defined by the two formation rules alone and are therefore distinction-only.

Theorem 5.3 (Orbit arithmetic). *The structure $(\mathbb{N}_\delta, 0_\delta, S_\delta, \oplus, \otimes)$ satisfies, with distinction-only strength:*

- (a) *additive and multiplicative associativity, $(a \oplus b) \oplus c = a \oplus (b \oplus c)$ and $(a \otimes b) \otimes c = a \otimes (b \otimes c)$;*
- (b) *additive and multiplicative commutativity, $a \oplus b = b \oplus a$ and $a \otimes b = b \otimes a$;*
- (c) *distributivity, $a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c)$;*
- (d) *identities, $a \oplus 0_\delta = a$ and $a \otimes S_\delta(0_\delta) = a$;*
- (e) *additive cancellation, $a \oplus c = b \oplus c \Rightarrow a = b$;*
- (f) *display faithfulness, $\varphi(a \oplus b) = \varphi(a) + \varphi(b)$ and $\varphi(a \otimes b) = \varphi(a) \cdot \varphi(b)$.*

Proof. Each item is proved by structural induction on $\mathbb{N}_\delta$, using only the recursive clauses (1) and (2). We give the additive arguments in full and indicate the rest, since all follow the same pattern.

First, a base lemma: $0_\delta \oplus n = n$ for all n . Induct on n . The case $n = 0_\delta$ is the base clause of addition. If $0_\delta \oplus m = m$, then $0_\delta \oplus S_\delta(m) = S_\delta(0_\delta \oplus m) = S_\delta(m)$. Next, a sliding lemma: $S_\delta(a) \oplus m = S_\delta(a \oplus m)$. Induct on m . For $m = 0_\delta$, both sides are $S_\delta(a)$. If $S_\delta(a) \oplus m = S_\delta(a \oplus m)$, then $S_\delta(a) \oplus S_\delta(m) = S_\delta(S_\delta(a) \oplus m) = S_\delta(S_\delta(a \oplus m)) = S_\delta(a \oplus S_\delta(m))$.

Additive associativity follows by induction on c . For $c = 0_\delta$, both sides are $a \oplus b$. The inductive step uses the recursive clause three times and the hypothesis once. Additive commutativity follows by induction on b , using the base lemma for $b = 0_\delta$ and the sliding lemma in the step: $a \oplus S_\delta(m) = S_\delta(a \oplus m) = S_\delta(m \oplus a) = S_\delta(m) \oplus a$. Additive cancellation follows by induction on c : the case $c = 0_\delta$ is immediate, and the step uses the injectivity of S_δ , which holds because S_δ is a formation rule with no identifications. Multiplicative associativity, multiplicative commutativity, and distributivity follow by the same method, each reducing to the additive laws already established. Display faithfulness is immediate from the recursive clauses and the recursive definition of φ . $\square$

We also fix the order on orbits, which we use throughout the sign and ratio constructions.

Definition 5.4 (Order and truncated subtraction on $\mathbb{N}_\delta$). The relation $\leq_\delta$ on $\mathbb{N}_\delta$ is generated by $0_\delta \leq_\delta n$ for all n , and $S_\delta(m) \leq_\delta S_\delta(n)$ if and only if $m \leq_\delta n$. Truncated subtraction $\ominus$ is defined by $n \ominus 0_\delta = n$, $0_\delta \ominus S_\delta(m) = 0_\delta$, and $S_\delta(n) \ominus S_\delta(m) = n \ominus m$.

Proposition 5.5 (The order is a decidable total order). *The relation $\leq_\delta$ is reflexive, antisymmetric, transitive, and total, and for any two orbits m, n the question whether $m \leq_\delta n$ is decided by a terminating recursion. Moreover $\leq_\delta$ is compatible with addition: $m \leq_\delta n$ if and only if $m \oplus c \leq_\delta n \oplus c$.*

Proof. Each property is proved by induction following the recursive clauses of Definition 5.4. The comparison recursion strips one successor from each argument at each step and halts when one argument reaches 0_δ , so it terminates after a number of steps equal to the smaller length. Because the recursion returns a definite yes or no for every input, the order is decidable without any appeal to excluded middle. Compatibility with addition follows by induction on c . $\square$

The decidability claim in Proposition 5.5 is the first concrete payoff of the budget discipline. In a classical setting one would assert that $m \leq_\delta n$ or $n <_\delta m$ by the law of excluded middle and move on. Here the disjunction is not asserted; it is computed. The comparison is a finite procedure on finite records, so the trichotomy holds constructively.

6 Sign as balance: the integers

The natural numbers have no sign. To subtract, one needs a direction. The usual construction defines an integer as an equivalence class of pairs of naturals under the relation $(a, b) \sim (c, d)$ exactly when $a + d = b + c$. The construction is correct, but the reason an integer points in a direction is invisible in it; one must know the convention that the first coordinate is the positive part. In the present setting, sign is balance, and the direction is visible in the object.

Definition 6.1 (Signed orbit and balanced equality). A *signed orbit* is an ordered pair $(p, n) \in \mathbb{N}_\delta \times \mathbb{N}_\delta$, where p is the positive part and n the negative part. Two signed orbits (a_+, a_-) and (b_+, b_-) are *balanced-equal*, written $(a_+, a_-) \approx (b_+, b_-)$, when

$$a_+ \oplus b_- = b_+ \oplus a_-$$

holds in $\mathbb{N}_\delta$. The *display* of a signed orbit is $\psi(p, n) = \varphi(p) - \varphi(n) \in \mathbb{Z}$.

The balance condition is the cross-addition condition, written entirely inside orbit arithmetic. It compares the two pairs without ever subtracting and without ever mentioning a negative number, which is exactly what one wants at the level where negative numbers are being constructed rather than assumed. For example, the integer -1 is any signed orbit (p, n) whose negative part exceeds its positive part by one performed distinction, such as $(0_\delta, S_\delta(0_\delta))$ or $(S_\delta(0_\delta), S_\delta(S_\delta(0_\delta)))$, and balance identifies all of them.

Definition 6.2 (The integers $\mathbb{Z}_\delta$). Set $\mathbb{Z}_\delta = (\mathbb{N}_\delta \times \mathbb{N}_\delta)/\approx$, the quotient of signed orbits by balanced equality. Operations descend from representatives:

$$[(a_+, a_-)] + [(b_+, b_-)] = [(a_+ \oplus b_+, a_- \oplus b_-)], \quad (3)$$

$$-[(a_+, a_-)] = [(a_-, a_+)], \quad (4)$$

$$[(a_+, a_-)] \cdot [(b_+, b_-)] = [(a_+ \otimes b_+ \oplus a_- \otimes b_-, a_+ \otimes b_- \oplus a_- \otimes b_+)]. \quad (5)$$

Negation swaps the positive and negative parts: it reverses direction, which is exactly what a sign change should mean for a balance. Addition adds the two positive parts and the two negative parts separately. Multiplication follows the familiar sign rule, written so that the positive part of a product collects the two like-signed cross terms and the negative part collects the two opposite-signed cross terms.

Theorem 6.3 (The integers form a ring isomorphic to $\mathbb{Z}$). *With distinction-only strength: the relation $\approx$ is an equivalence relation; the operations (3) through (5) are well-defined on the quotient; $\mathbb{Z}_\delta$ with these operations is a commutative ring with additive identity $[(0_\delta, 0_\delta)]$ and multiplicative identity $[(S_\delta(0_\delta), 0_\delta)]$; and the display map $\Psi: \mathbb{Z}_\delta \rightarrow \mathbb{Z}$, $\Psi([(p, n)]) = \varphi(p) - \varphi(n)$, is a ring isomorphism.*

Proof. That $\approx$ is an equivalence relation follows from the additive laws of Theorem 5.3. Reflexivity is $a_+ \oplus a_- = a_+ \oplus a_-$. Symmetry is immediate. For transitivity, suppose $a_+ \oplus b_- = b_+ \oplus a_-$ and $b_+ \oplus c_- = c_+ \oplus b_-$. Adding the two equalities and using commutativity and associativity gives $(a_+ \oplus c_-) \oplus (b_+ \oplus b_-) = (c_+ \oplus a_-) \oplus (b_+ \oplus b_-)$, and additive cancellation, Theorem 5.3(e), removes the common term $b_+ \oplus b_-$ to yield $a_+ \oplus c_- = c_+ \oplus a_-$.

Well-definedness of the operations is the standard cross-addition computation. For addition, if $(a_+, a_-) \approx (a'_+, a'_-)$ and $(b_+, b_-) \approx (b'_+, b'_-)$, adding the two balance equalities and rearranging gives the balance equality for the sums. For negation, swapping parts turns one balance equality into the other. For multiplication, one expands both sides using distributivity and the balance hypotheses; the computation is finite and uses only Theorem 5.3. The ring axioms transfer from $\mathbb{N}_\delta$ in the same way: each is verified on representatives and respects balance.

The display map Ψ is well-defined because balanced orbits have equal display: $a_+ \oplus b_- = b_+ \oplus a_-$ implies, under display faithfulness, $\varphi(a_+) - \varphi(a_-) = \varphi(b_+) - \varphi(b_-)$. It is injective

because equal display forces balance, again by faithfulness. It is surjective because each integer $k \geq 0$ is $\Psi([\varphi^{-1}(k), 0_\delta])$ and each $k < 0$ is $\Psi([0_\delta, \varphi^{-1}(-k)])$, where $\varphi^{-1}(k)$ is the orbit of length k . It preserves the operations by direct computation. Hence Ψ is a ring isomorphism. $\square$

The integers also carry order, comparison, and absolute value, all native to the signed-orbit representation.

Theorem 6.4 (Order, comparison, and absolute value on $\mathbb{Z}_\delta$). *With distinction-only strength, $\mathbb{Z}_\delta$ carries:*

- (a) *a total order, $[(a_+, a_-)] \leq [(b_+, b_-)]$ if and only if $a_+ \oplus b_- \leq_\delta b_+ \oplus a_-$, well-defined on the quotient and compatible with addition on either side;*
- (b) *a three-way comparison selector whose outputs are equivalent to strict less-than, balanced equality, and strict greater-than, computed by a terminating recursion;*
- (c) *an absolute value $||[(a_+, a_-)]|| = (a_+ \ominus a_-) \oplus (a_- \ominus a_+)$, invariant under balanced equality and under negation.*

The comparison selector and absolute value are defined inside the signed-orbit representation; the display map is used only to certify them, not to define them.

Proof. For (a), well-definedness on the quotient follows by the same cross-addition argument as in Theorem 6.3, replacing equalities with the order $\leq_\delta$ of Proposition 5.5, which is compatible with addition. Totality and transitivity transfer from $\leq_\delta$. Compatibility with integer addition follows from compatibility of $\leq_\delta$ with orbit addition. For (b), the selector compares $a_+ \oplus b_-$ with $b_+ \oplus a_-$ using the terminating comparison of Proposition 5.5 and returns one of three outcomes; the equivalence of those outcomes with strict order and balance is read off from the definition of $\leq_\delta$. For (c), the expression $(a_+ \ominus a_-) \oplus (a_- \ominus a_+)$ equals the larger part minus the smaller in orbit terms, since one of the two truncated differences is 0_δ ; invariance under balance and under the swap that defines negation is a finite check using Theorem 5.3. The transport identities $\text{cmp}(c + a, c + b) = \text{cmp}(a, b)$ and $\text{cmp}(-b, -a) = \text{cmp}(a, b)$ follow from the additive compatibility and the swap symmetry. $\square$

7 Ratio as cross-comparison: the rational field

A ratio compares two integer counts. The usual construction defines a rational as an equivalence class of pairs under cross-multiplication. We use the same relation, expressed entirely in the integer and orbit arithmetic already built.

Definition 7.1 (Ratio orbit and $\mathbb{Q}_\delta$). A *ratio orbit* is a pair (a, b) in which $a \in \mathbb{Z}_\delta$ is the numerator and $b \in \mathbb{N}_\delta \setminus \{0_\delta\}$ is a nonzero denominator length. Write $\hat{b} = [(b, 0_\delta)] \in \mathbb{Z}_\delta$ for the integer that lifts the denominator. Two ratio orbits are *cross-equal*, written $(a, b) \sim_\times (c, d)$, when

$$a \cdot \hat{d} = c \cdot \hat{b}$$

holds in $\mathbb{Z}_\delta$. Set $\mathbb{Q}_\delta = \{\text{ratio orbits}\} / \sim_\times$.

For instance, the fractions $\frac{2}{3}$ and $\frac{4}{6}$ are ratio orbits with numerators $[(2_\delta, 0_\delta)]$ and $[(4_\delta, 0_\delta)]$ and denominators 3_δ and 6_δ , where k_δ denotes the orbit of length k . Cross-equality checks $2_\delta \otimes 6_\delta = 4_\delta \otimes 3_\delta$, that is $12_\delta = 12_\delta$, using orbit multiplication alone. No prior notion of fraction enters.

Theorem 7.2 (The rationals form an ordered field isomorphic to $\mathbb{Q}$). *With distinction-only strength: $\sim_\times$ is an equivalence relation; the operations*

$$[(a, b)] + [(c, d)] = [(a \cdot \hat{d} + c \cdot \hat{b}, b \otimes d)], \quad (6)$$

$$[(a, b)] \cdot [(c, d)] = [(a \cdot c, b \otimes d)], \quad (7)$$

$$[(a, b)]^{-1} = [(\varepsilon_\delta(a) \hat{b}, |a|)] \quad (a \neq 0) \quad (8)$$

are well-defined on the quotient, where $\varepsilon_\delta(a)$ is the internal orientation of a , equal to $[(S_\delta(0_\delta), 0_\delta)]$ when the signed-orbit nonnegative test of Theorem 6.4 succeeds and $[(0_\delta, S_\delta(0_\delta))]$ otherwise, and $|a| \in \mathbb{N}_\delta$ is the orbit length underlying the absolute value; under these operations $\mathbb{Q}_\delta$ is an ordered field; and the display map $\Phi: \mathbb{Q}_\delta \rightarrow \mathbb{Q}$, $\Phi([(a, b)]) = \psi(a)/\varphi(b)$, is an isomorphism of ordered fields.

Proof. That $\sim_\times$ is an equivalence relation follows from the commutativity and associativity of integer multiplication, Theorem 6.3, together with the fact that the lifted denominator $\hat{b}$ is never a zero divisor because $b \neq 0_\delta$. Reflexivity and symmetry are immediate; transitivity uses cancellation of the nonzero common denominator factor, which is available in $\mathbb{Z}_\delta$ since $\mathbb{Z}_\delta$ is an integral domain by Theorem 6.3.

Well-definedness of (6) through (8) is the standard cross-multiplication computation, carried out in $\mathbb{Z}_\delta$. If $(a, b) \sim_\times (a', b')$, then $a \cdot \hat{b}' = a' \cdot \hat{b}$, and substituting into each operation and expanding with the ring laws preserves cross-equality of the results. The reciprocal deserves a word: its denominator is the internal absolute value $|a|$, a genuine orbit length and hence a legal denominator because $a \neq 0$ forces $|a| \neq 0_\delta$, and its numerator is the orientation of a times the lifted old denominator. The orientation is selected by the internal comparison of Theorem 6.4, not by any external sign function, so the reciprocal is defined inside the construction. One checks directly that $[(a, b)] \cdot [(a, b)]^{-1}$ cross-equals the unit ratio $[(S_\delta(0_\delta), 0_\delta)]$.

The field axioms transfer from the ring laws of $\mathbb{Z}_\delta$ by computation on representatives. The order on $\mathbb{Q}_\delta$ is induced from the integer order by the rule that $[(a, b)] \leq [(c, d)]$ when $a \cdot \hat{d} \leq c \cdot \hat{b}$ after normalizing denominators to be positive, which is legitimate because every ratio orbit has a representative with positive denominator length. Finally Φ is well-defined and injective because cross-equality matches the standard cross-multiplication criterion for equality of fractions under display faithfulness, surjective because every fraction p/q in lowest terms is the display of $[(\Psi^{-1}(p), \varphi^{-1}(q))]$, and operation- and order-preserving by direct computation. Hence Φ is an isomorphism of ordered fields. $\square$

8 The permission surface: divisibility to greatest common divisor

With the number tower in place, the elementary number-theoretic surface follows without any increase in proof budget. We collect the constructions in one statement and then comment on what each one requires.

Theorem 8.1 (Elementary permission surface). *The following are constructed with distinction-only strength, each decidable by a terminating procedure on finite orbit data.*

- (a) **Divisibility.** For $a, b \in \mathbb{N}_\delta$, write $a \mid b$ when there is $c \in \mathbb{N}_\delta$ with $b = a \otimes c$. Divisibility is decidable.
- (b) **Units and primality.** The only multiplicative unit of $\mathbb{N}_\delta$ is $S_\delta(0_\delta)$. An orbit p is prime when $p \neq 0_\delta$, $p \neq S_\delta(0_\delta)$, and every factorization $p = a \otimes b$ has $a = S_\delta(0_\delta)$ or $b = S_\delta(0_\delta)$. Primality is decidable.
- (c) **Euclidean division.** For $a \in \mathbb{N}_\delta$ and $b \neq 0_\delta$, there exist unique $q, r \in \mathbb{N}_\delta$ with $a = (b \otimes q) \oplus r$ and $r <_\delta b$.

- (d) **Greatest common divisor.** *The Euclidean algorithm on orbits produces $\gcd_\delta(a, b)$, the largest orbit dividing both. Two orbits are coprime when $\gcd_\delta(a, b) = S_\delta(0_\delta)$.*

Each structure has a faithful display: φ carries divisibility, primality, Euclidean quotient and remainder, and greatest common divisor on $\mathbb{N}_\delta$ to the corresponding facts on $\mathbb{N}$.

Proof. For (a), the search for a witness c with $b = a \otimes c$ may be bounded: if $a \neq 0_\delta$, any witness has length at most $\varphi(b)$, so the search is finite; if $a = 0_\delta$, then $a \mid b$ exactly when $b = 0_\delta$. Either way the question is decided in finitely many steps. For (b), that $S_\delta(0_\delta)$ is the only unit follows because $a \otimes b = S_\delta(0_\delta)$ forces both factors to have length one by display faithfulness; primality is then a finite search over the candidate factors of p , all of which have length at most $\varphi(p)$.

For (c), existence is by induction on a using repeated subtraction: if $a <_\delta b$ then $(q, r) = (0_\delta, a)$; otherwise apply the hypothesis to $a \ominus b$, which is strictly smaller, and increment the quotient. The recursion terminates because the first argument strictly decreases at each step and $\mathbb{N}_\delta$ is well-founded under $<_\delta$, a fact that itself follows by induction without any appeal to a least-number principle phrased classically. Uniqueness follows from cancellation, Theorem 5.3(e): if $(b \otimes q) \oplus r = (b \otimes q') \oplus r'$ with $r, r' <_\delta b$, then $q = q'$ and $r = r'$, since a difference of one in the quotient would force a remainder of size at least b .

For (d), the Euclidean algorithm replaces the pair (a, b) by $(b, a \bmod b)$ until the second entry is 0_δ ; termination is guaranteed because the second entry strictly decreases under $<_\delta$. The output divides both inputs and is divisible by every common divisor, which gives the greatest-common-divisor property. Coprimality is then the decidable condition that the output is $S_\delta(0_\delta)$. Display faithfulness in each case is inherited from Theorem 5.3: φ preserves products, hence divisibility, hence the derived notions. $\square$

It is worth pausing on what these proofs did not need. The proof of primality in Theorem 8.1(b) used orbit multiplication and the unit, and nothing else: no set membership, no equivalence-class encoding, and no completed infinity. The prime orbits are exactly the orbits that cannot be written as a product of two non-unit orbits, and that property is visible in the object at the point of construction. The same holds down the list. Order on the integers is balance comparison. Euclidean division is repeated subtraction that terminates because orbits are finite records. The greatest common divisor is the ordinary Euclidean algorithm running on orbit arithmetic rather than on an encoding of arithmetic inside a foreign primitive. In each case the structural reason for the result is present in the construction, not recovered afterward by decoding.

9 What the budget buys

The constructions above are isomorphic to the familiar number systems, so the contribution is not a new arithmetic. It is the account of what produced the arithmetic. Three points make the account precise.

9.1 No completed infinity

We never posited an infinite set. Each orbit is a finite record, generated by zero and successor, and each theorem is proved by structural induction over such records. When we wrote $\mathbb{N}_\delta$, $\mathbb{Z}_\delta$, or $\mathbb{Q}_\delta$ and quantified over their elements, we quantified over collections generated by formation rules, in the same way one quantifies over the natural numbers themselves. A reader trained in set theory may ask whether the inductive collection $\mathbb{N}_\delta$ is not itself an instance of the axiom of infinity in disguise. The honest answer is that the inductive generation of $\mathbb{N}_\delta$ is exactly as strong as accepting the natural numbers as a generated collection, and no stronger; what we avoid is the separate set-theoretic axiom that asserts a completed infinite set as a new object with its own existence postulate, and the constructions never require treating $\mathbb{N}_\delta$ as a completed

totality rather than as a rule for generating its members. The distinction matters at the next layer. Completing $\mathbb{Q}_\delta$ to the continuum does require a genuinely new commitment, and that is precisely the boundary marked in Section 10.

9.2 No choice

Every object the constructions need is produced explicitly. Representatives are not selected from equivalence classes by a choice function; instead, all constructions are defined on representatives and shown to respect the equivalence, so they descend to the quotient by the principle of Definition 3.2. Witnesses for divisibility, quotients and remainders for Euclidean division, and the output of the Euclidean algorithm are all produced by terminating procedures, not chosen from a family of candidates. There is therefore no point at which the axiom of choice could enter.

9.3 No excluded middle

Every predicate the constructions rely on is decidable. Order on orbits, equality of orbits, divisibility, primality, and coprimality are each computed by a recursion that halts and returns a definite answer on finite data. We never needed to assert a disjunction without being able to compute which disjunct holds. The order trichotomy, in particular, is a computed comparison rather than a classical case split. This is why every result carries the distinction-only tag rather than a tag that admits classical logic.

9.4 The display maps are instruments, not ingredients

The maps φ , ψ , Ψ , and Φ translate the distinction-native objects into the standard naturals, integers, and rationals, and we proved that the relevant ones are isomorphisms. Their role is conservativity: they certify that the construction recovers exactly the familiar objects and produces nothing stranger. They are not used to define the native operations. Orbit addition, signed-orbit balance, ratio cross-equality, and the internal orientation that defines the reciprocal are all given by rules on the native representation, and the display maps are applied afterward to verify that those rules behave as the standard objects do. A reader who deleted the display maps entirely would still have a complete, self-contained number tower; the maps are there so that the reader can check the tower against the arithmetic already known.

10 Where the forcing stops

The constructions of this paper end at the rational field, and that endpoint is deliberate. Everything up to and including $\mathbb{Q}_\delta$ is forced by distinction and finite repetition, with the budget held fixed at the distinction-only level. The next natural step, completing $\mathbb{Q}_\delta$ into the continuum so that limits and suprema exist, is not forced by distinction, and the reason is structural rather than a matter of effort. Distinction is a generative act that proceeds one step at a time, so everything it can produce is reachable by a finite or step-indexed generation, and the collection of all such products is countable. The continuum is uncountable. A countable generative process cannot produce an uncountable object, so the real line, and the completeness principle that singles it out among ordered fields, is an addition to distinction rather than a consequence of it.

We do not prove that claim here; it is the subject of the companion paper, which establishes the non-forcing of the continuum with two independent arguments and draws the consequences for the wider program. The point relevant to the present paper is only this: the arithmetic tower is self-contained and complete at the distinction-only level, and it is honest about its own ceiling. The boundary between what distinction forces and what it merely permits as a further commitment falls exactly between the rational field and its completion. Naming that boundary

precisely is part of what it means to take a single primitive seriously, and it is the reason the arithmetic layer can be presented on its own, as a finished result, without overstating what the primitive reaches.

11 Conclusion

From one act, the act of distinction, we built finite traces, a sameness relation and its consistent companion difference relation, and a quotient discipline. On that kernel we constructed the natural numbers as distinction orbits, the integers as balanced pairs of orbit lengths, and the rationals as cross-compared ratios, and we proved the complete elementary arithmetic of each: the algebraic laws, a decidable total order, absolute value, divisibility, primality, Euclidean division with unique quotient and remainder, greatest common divisors, and the ordered-field axioms. Every result was established with the same minimal proof budget, using no completed infinity, no choice, and no classical case analysis, and we kept an explicit ledger of that budget so the claim could be checked at each step. The integers and rationals so constructed are isomorphic to the standard ones under faithful display maps, which we use to certify the construction rather than to power it.

The value of the result is the audit, not a new arithmetic. It shows that the elementary number tower is not a convention layered on top of a foundation but a structure that the single act of distinction permits and then forces. It also shows, by stopping where it does, exactly how far that forcing reaches: up to the rational field and no further without a new commitment. Mathematics, at this layer, is what distinction permits. The next paper takes up what happens when one asks for more, and shows that the continuum is the first thing distinction cannot supply on its own.

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